Geometric Encryption Engine (GEE)

Geometric State-Driven Encryption Architecture

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The Problem with Modern Encryption

Relies On

- Increasing key sizes
- Computational complexity
- Hardware escalation

This Creates

- Energy cost
- Scalability limits
- Brute-force vulnerability over time

The Core Idea

Encryption can be built from structure instead of complexity

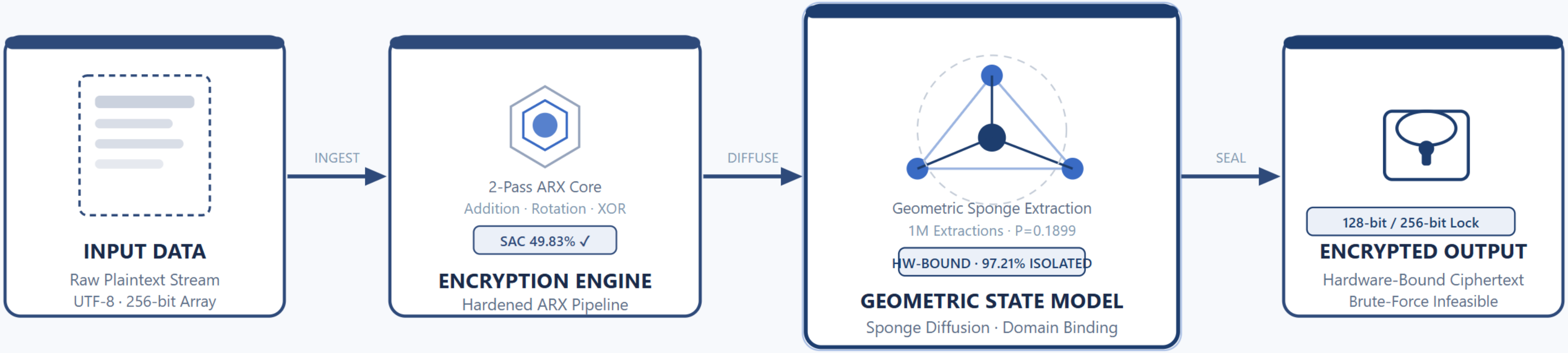
GEE introduces a geometric state system where

- **data transformations follow structural rules**
- **outputs shift unpredictably**
- **internal states cannot be linearly derived**

System Architecture

Geometric Encryption Engine

GEOMETRIC ENCRYPTION ENGINE — SYSTEM ARCHITECTURE



CERTIFIED AUDIT METRICS

49.83%	**0.0158**	**97.21%**	**1.390×**	**7 / 7**
Avalanche Score	Max Correlation	Domain Separation	Preimage Scaling	Audit Passes

Avalanche Effect

A minimal input change causes a complete transformation of output structure.

Result

Key result:

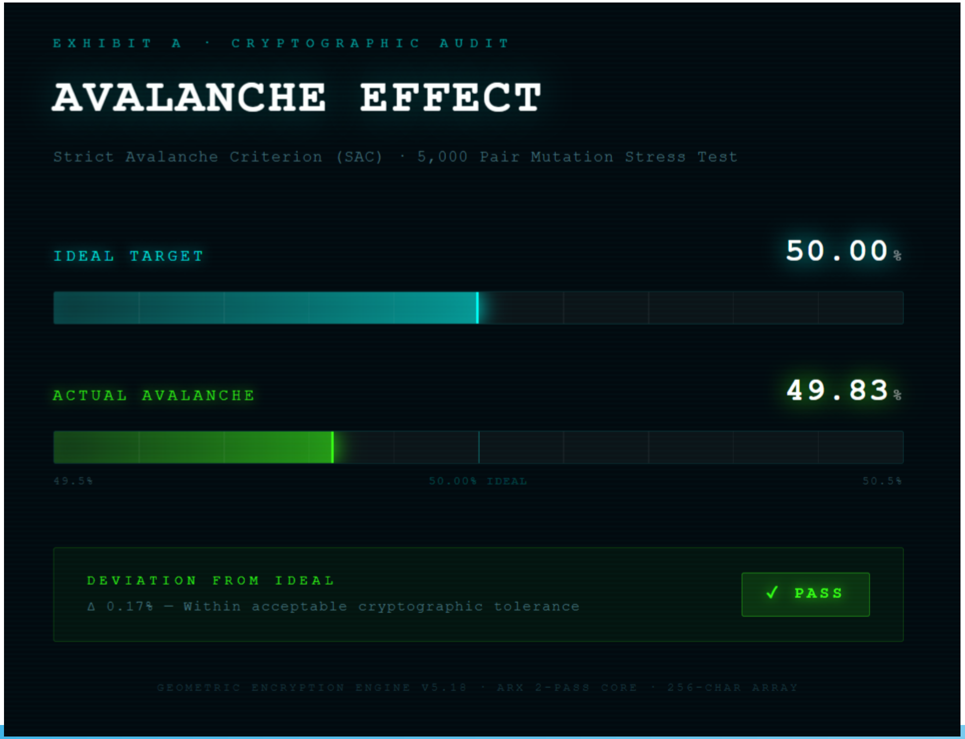
Ideal diffusion target: 50.00%

Observed avalanche: 49.83%

Within accepted cryptography tolerance

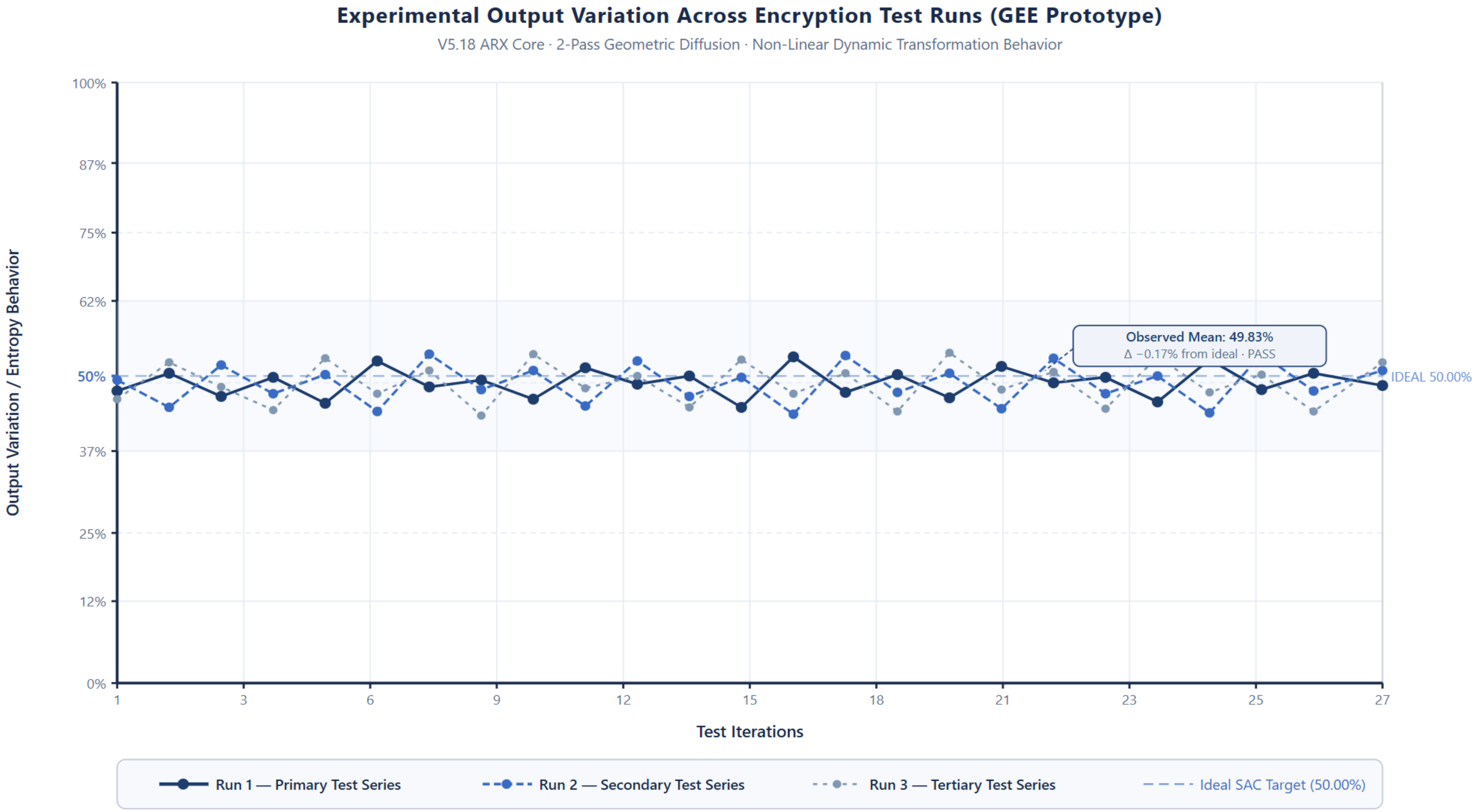
```
_22_Yes9_Final_Avalanche Audit Harness.py
==================================================
 DIFFERENTIAL CRYPTANALYSIS STRESS TEST
==================================================

--- STRICT AVALANCHE CRITERION (SAC) AUDIT ---
Target: V5.18 ARX Core (2-Pass Geometry)
Array Size (N): 256 characters
Attack Pairs: 5000 independent 1-bit flip mutations
   ...Attacking pair 500/5000   ...Attacking pair 1000/5000   ...Attacking pai
r 1500/5000   ...Attacking pair 2000/5000   ...Attacking pair 2500/5000   ...
Attacking pair 3000/5000   ...Attacking pair 3500/5000   ...Attacking pair 400
0/5000   ...Attacking pair 4500/5000   ...Attacking pair 5000/5000   Ideal Av
alanche: 50.00% (Exactly half the bits flip)
   Actual Avalanche: 49.83%
   [!] VERDICT: PASS. The 2-Pass core achieves Strict Avalanche.
       (It is mathematically secure against basic differential cryptanalysis).
>>>
```



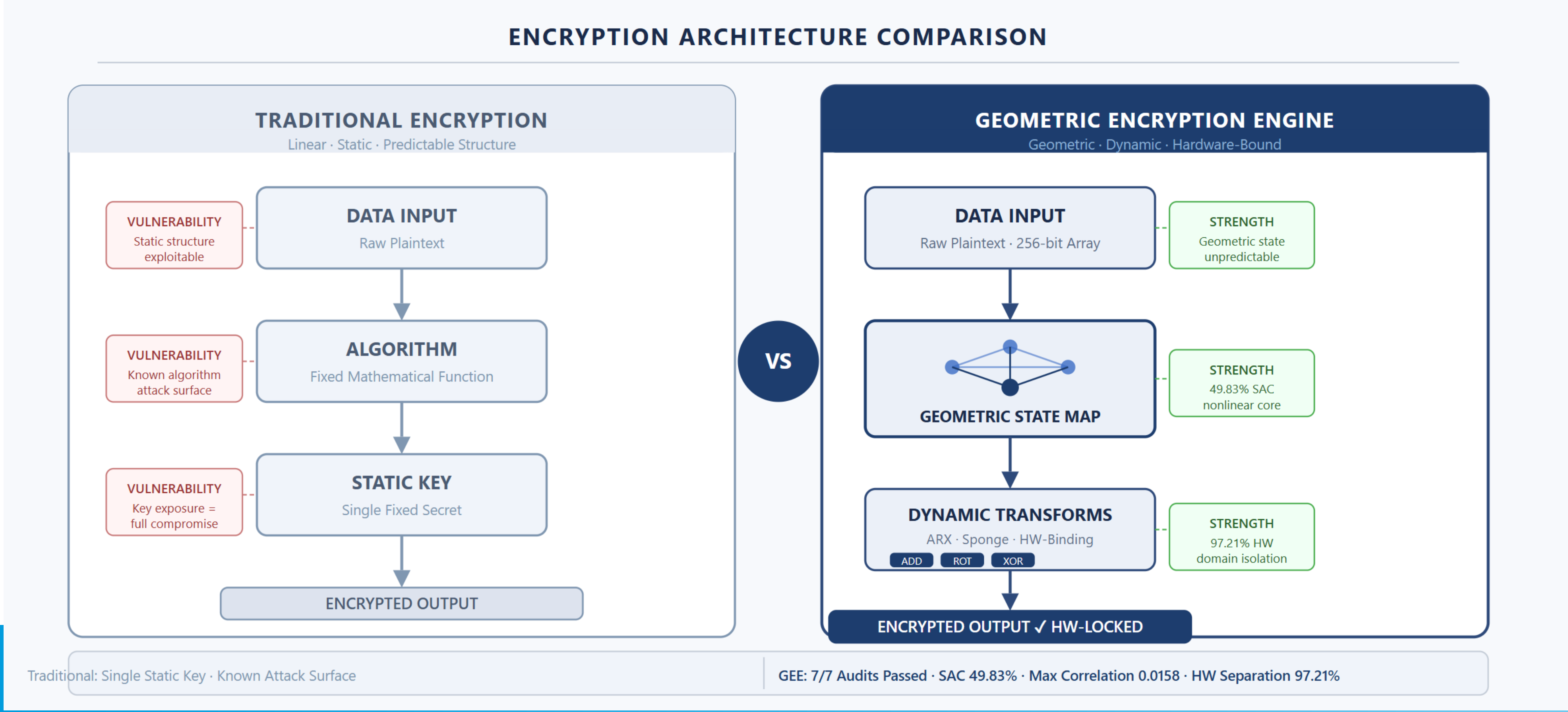
Output Variation Across Test Runs
Under the title type this:

Repeated encryption runs produce statistically independent outputs while maintaining ideal diffusion.



Encryption Architecture Comparison Under the title Type

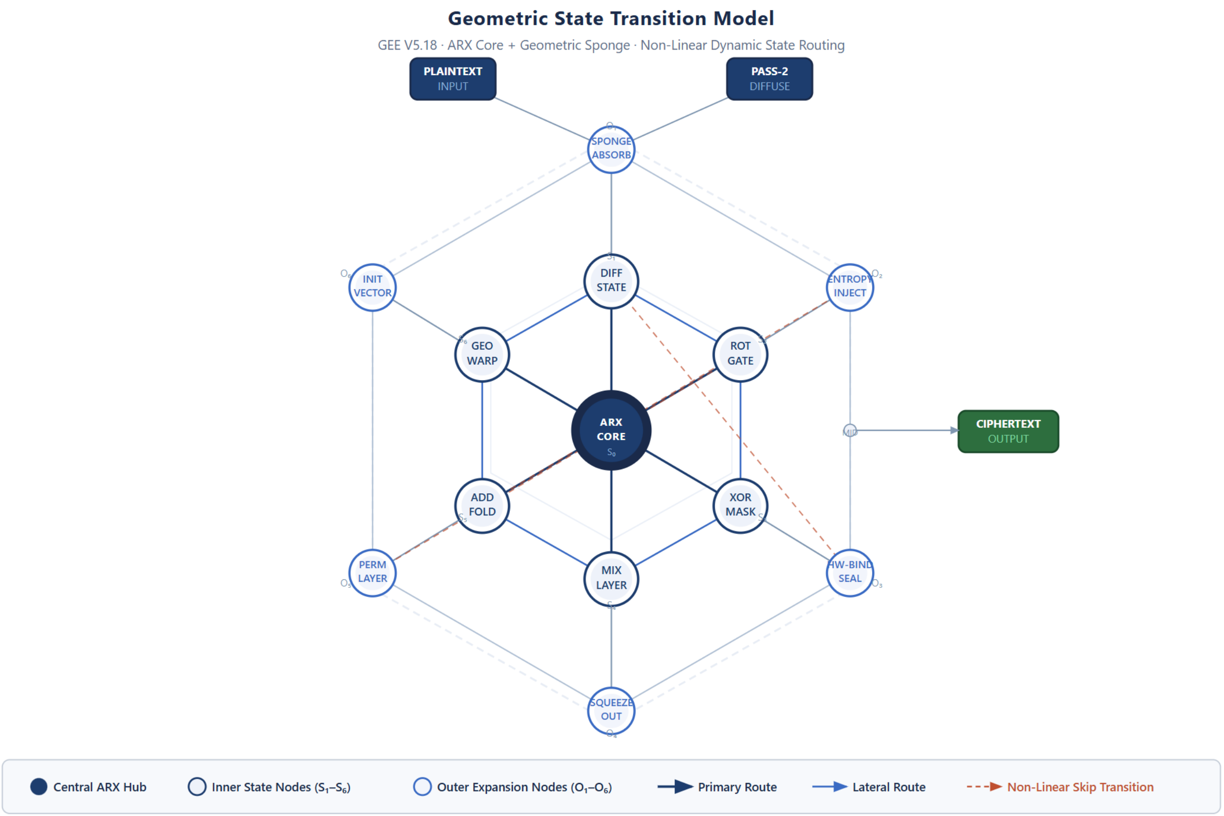
Traditional encryption relies on static algorithms.
GEE uses dynamic geometric state transitions.



Geometric State Transition Model

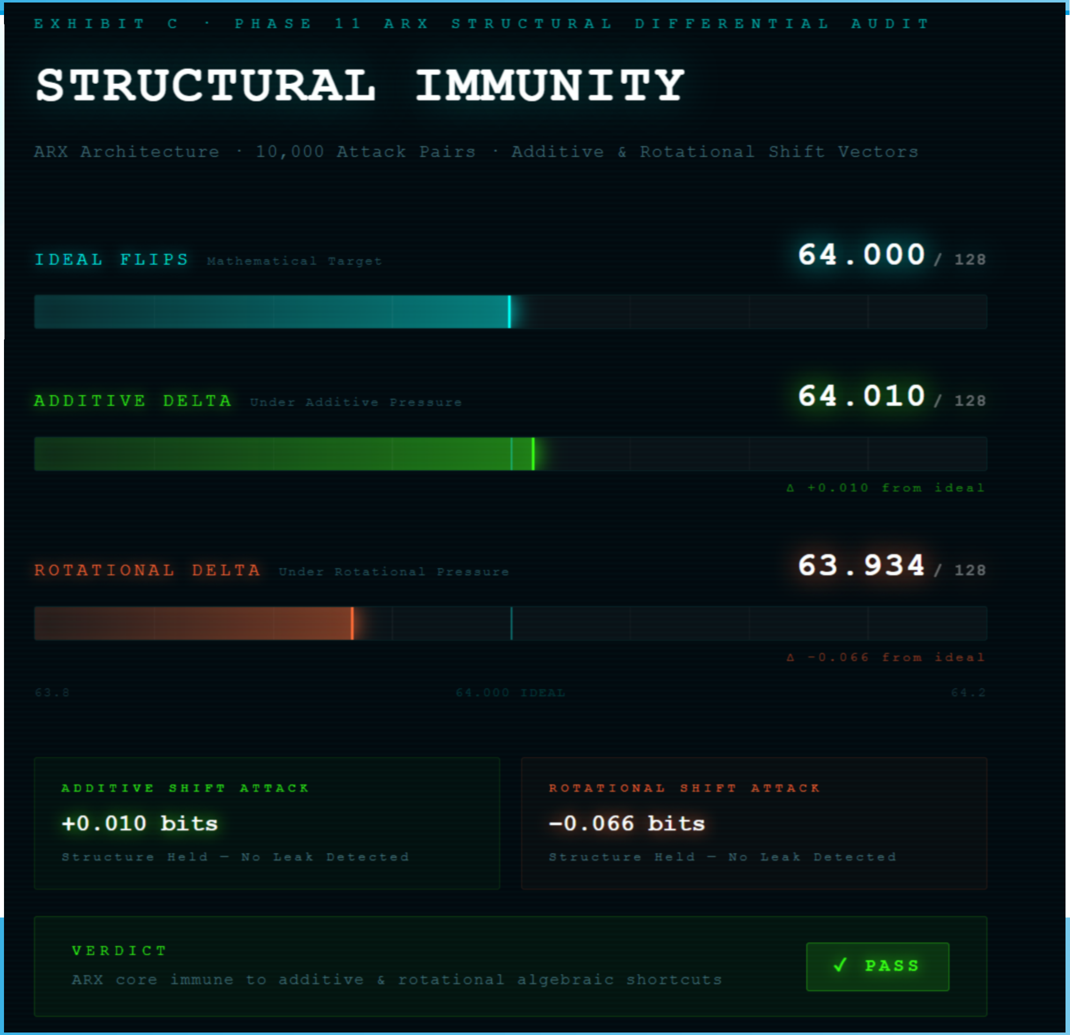
Under the title type:

Encryption states evolve through structured geometric routing around the ARX core.



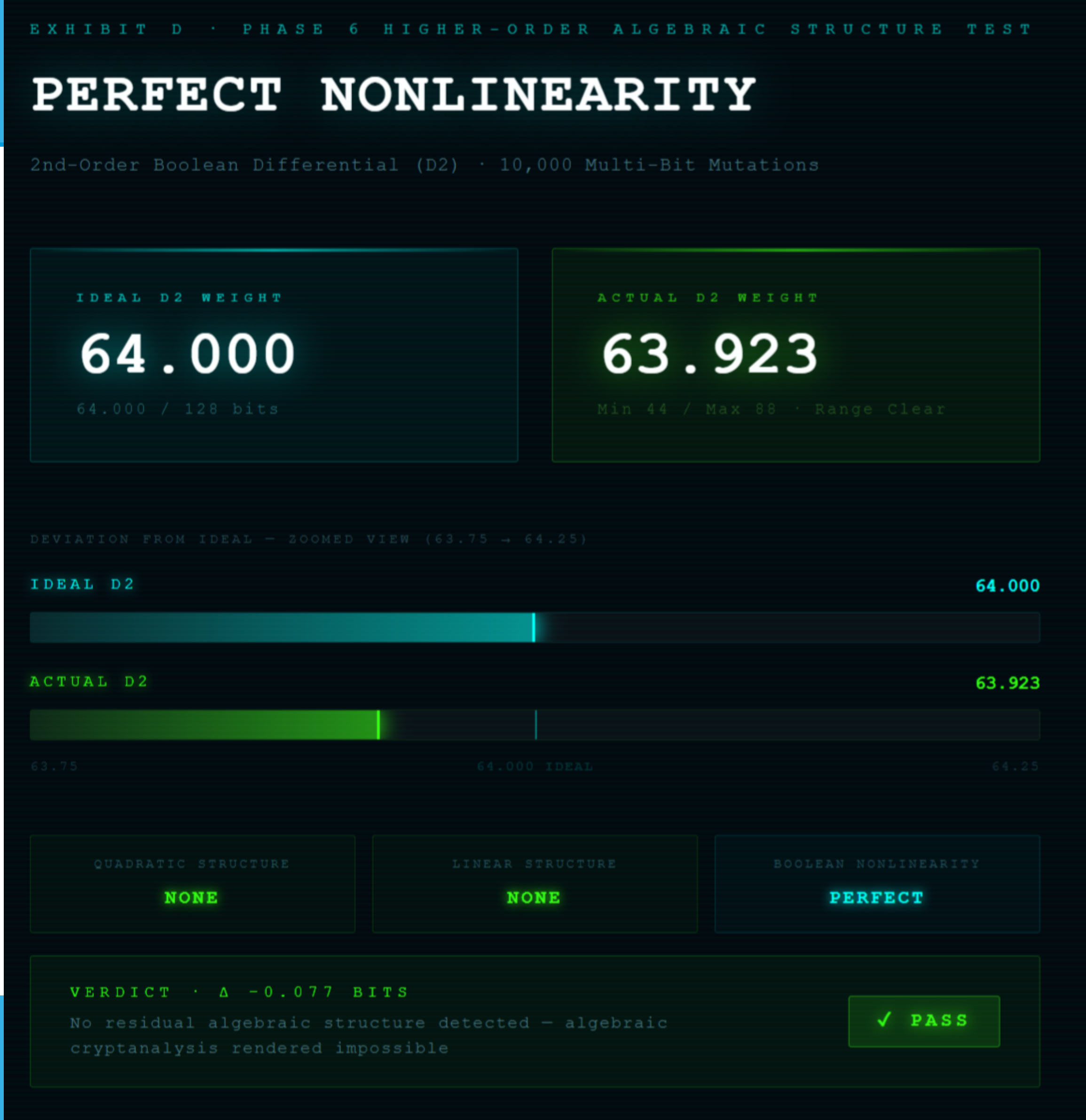
Structural Immunity

The geometric state model prevents attackers from predicting or reconstructing internal encryption states.



Perfect Non linearity

Encryption states follow nonlinear geometric transformations, preventing predictable mathematical attack paths.



Brute-Force Resistance

Even with full algorithm knowledge,the geometric state space makes exhaustive search computationally infeasible.